

CHAPTER 5: Interaction Design Basics

5.1 Introduction

Interaction Design: Designing the **interaction between users and technology**, not just the system itself.

- Design interventions that improve existing situations.
- Consider how technology changes the way people work and live.

The lesson

Interaction design is not just about the artifact (a device or program); it is about understanding and choosing **how that artifact will affect the way people work**.

The artifacts we give people also include manuals, tutorials, and online help systems.

5.2 What Is Design?

Design = **achieving goals within constraints**.

Goals

- **What the design should achieve.**
- Who it is for.

Constraints

- Cost
- Time
- Standards
- Safety

- Technology limits

Trade-Offs

- **Balancing conflicting goals** and **constraints**.

Important Point : Good design is about **managing trade-offs**, not optimizing only one goal.

5.2.1 The Golden Rule of Design

Golden rule → **Understand your materials.**

- **Understand computers** — their limitations, capacities, tools, and platforms.
- **Understand people** — their psychological and social aspects, and human error.

5.2.2 To Err Is Human

Humans naturally make mistakes.

Types of Mistakes

- Slips
- Errors
- Omissions

Designer's Responsibility

- **Reduce mistakes.**
- Minimize consequences of mistakes.

Important Point

- Do not blame users.
- **Design systems that tolerate human error.**

5.2.3 The Central Message – The User

Core of interaction design

Put the **user first**, keep the **user in the centre**, and **remember the user at the end**.

5.3 The Process of Design

Flow of the process

Requirements ('What is wanted') – gathered via interviews and ethnography; what is there vs. what is wanted.

Analysis – using scenarios and task analysis to order findings.

Design – guided by guidelines and principles; recorded with dialog notations.

Prototype – evaluated using heuristics and evaluation; the loop returns to Analysis/Design.

Implement and deploy – needing precise specification, architectures, documentation, and help.

The Four Phases (plus iteration)

1. Requirements: What is wanted?

- Interviews + Observation + Video recording + Ethnography

2. Analysis: Organize collected information.

- Task analysis
- Task models
- Scenarios

3. Design: Decide how the system will work.

- Guidelines + Principles + Models + Design notations

4. Iteration & Prototyping

- Build prototype.
- Evaluate.
- Improve.
- Repeat.

5. Implementation & Deployment

- Coding
- Documentation
- Deployment
- Maintenance

The Reality of Limited Time

Trade-off between design time and design quality

Not every usability problem should be fixed.

Balance quality, cost, and time.

5.4 User Focus

The commandment> **Know your users**

Stakeholder – Anyone affected directly or indirectly by the system.

- Users
- Managers
- Customers
- Auditors

Getting to Know Users

Who Are They?

- Age
- Experience
- Skills
- Goals

Talk to Them

Participatory Design: Users participate throughout design.

- Better understanding.
- Better acceptance.
- User ownership.

Watch Them: Observe what users actually do.

- People often cannot accurately explain their actions.

Looking at Artifacts

Artifacts → Objects people use.

- Documents
- Sticky notes
- Tools

Purpose: Understand work practices.

Cultural Probes

Small packs used to collect information from users in their natural environment.

- Camera, Recorder, Notes

Discuss Observations

- Observation Tells: **WHAT users do.**
- Discussion Tells: **WHY they do it.**

Use Your Imagination

Method Acting: Imagine what the user would do, not what you would do.

Persona: Detailed fictional user representing a user group.

- Help designers think like real users.

5.5 Scenarios

story describing **interaction between user and system.**

Purpose

- Understand usage.
- Explore designs.
- Find problems.

Storyboards: Scenarios with sketches or screen mockups.

› Why Scenarios Are Useful

- Detailed Thinking: **Reveals hidden problems.**
- Communication: **Easy to explain designs.**
- Validation: **Check other models.**
- Dynamics: **Shows behavior over time.**

› Limitation

- Shows one path only.
- May miss alternative paths.

Scenarios as Reusable Resources

Uses

- Requirements gathering
- Design exploration
- Implementation checking
- Test case generation

5.6 Navigation Design

Help users **move through the system**.

- **Widgets** — Appropriate choice of widgets and the wording in menus and buttons help you know how to use them for a selection or action.
- **Screens or windows** — need to **find things** on the screen and **understand the logical grouping of buttons**.
- **Navigation within the application** — **what will happen when a button is pressed**, and **where you are in the interaction**.

Table 5.1 — Levels of interaction

PC application	Website	Physical device
Widgets	Form elements, tags and links	Buttons, dials, lights, displays
Screen design	Page design	Physical layout
Navigation design	Site structure	Main modes of device
Other apps and operating system	The web, browser, external links	The real world!

- **Local structure** — looking from one screen or page out.
- **Global structure** — the structure of the site and movement between screens.

5.6.1 Local Structure

Four things to check for any single screen / page / device state

1. **Knowing where you are.**
2. **Knowing what you can do.**
3. **Knowing where you are going — or what will happen.**
4. **Knowing where you've been — or what you've done.**

Breadcrumbs: **Show user's location within a hierarchy.**

Icons need labels

Icons alone are often unclear.

Use **labels** or **tooltips**.

Design Focus: Beware the big button trap

The big button trap

Public touchscreen systems have large buttons, yet users frequently hit the wrong screen and use 'back'/'home' to try again.

Each big button often has only one or two words plus an icon, in a large font (rightly, for poor eyesight).

Fix: there is usually room for additional explanation in a smaller font (the next level of labels, or a sentence). It looks less pretty but helps people actually find the information.

Modes

- **Same action** has **different meanings** in **different contexts**.
- Can cause errors.
- **Solution** → **Make modes visible**.

5.6.2 Global Structure — Hierarchical Organization

Hierarchy: Organizing screens into levels.

- Easy to understand.
- Logical organization.

Problem : (Different users think differently.

Deep Hierarchy

- **Many levels.**
- Harder navigation.

Broad Hierarchy

- **More options per level.**
- Often preferred.

Menu breadth — a common misuse of Miller's magic number

Miller's 7 ± 2 applies to WORKING MEMORY, not visual search — so the guideline that menu breadth should be about seven is a misuse.

5.6.3 Global Structure — Dialog

- **Pattern of interaction between user and system.**

Network diagrams

Scenarios give just one path. To describe a full system you must include **different paths and loops**. A simple way is a network diagram showing principal states/screens linked with arrows. It can represent

- Paths

- Branches
- Loops
- Screen transitions

5.6.4 Wider Still — The Environment

Systems do not exist alone.

Style Issues

- Follow platform standards.
- Maintain consistency.

Functional Issues

- File handling
- Data sharing
- Copy/Paste

Navigation Issues

- Links between applications.

5.7 Screen Design and Layout

A single screen must present information clearly and act as the locus for interaction.

- **Ask** — What is the user doing?
- **Think** — What information is required? What comparisons may the user need to make? In what order are things likely to be needed?
- **Design** — Form follows function — let the required interactions drive the layout.

5.7.1 Tools for Layout

Grouping and structure: Place related items together.

Order of groups and items: Arrange items in natural user order.

Decoration: Boxes, separating lines, font style, and text/background colours make grouping clear

Alignment: text “left-aligned”; numbers “right-aligned” / decimal aligned

White space : Separate / Create visual gaps.

5.7.2 User Action and Control

Entering information

⇒ **Good Form Design**

- Clear grouping.
- Logical order.
- Easy navigation.

Knowing what to do

Active Elements (expecting input or action) Must be clearly identifiable.

- Buttons
- Menus
- Input fields

Affordances

- Appearance suggests how something should be used.
- Button → Press

- Handle → Pull
- Affordances depend on user experience and culture.

5.7.3 Appropriate Appearance

Presenting information

- **Presentation** should match the **purpose**
- **Alphabetical List:** Good for finding names.
- **Date List:** Good for recent items.

Aesthetics and utility

- Beautiful interfaces are **not always usable**.
- **Balance appearance and usability.**

Making a mess of it: colour and 3D

Colour Guidelines

- Distracting + color blindness issues

Use colour **sparingly**; **do not rely on it to give information** — use it to **REINFORCE** other attributes.

3D effects

Excellent for **physical information** and certain **graphs/visualization**

> bad for text / pie charts

Localization / Internationalization

- **Adapting software** for different languages and cultures.
- **ISSUES**
 - Language
 - Layout

- Symbols
- Cultural meanings

5.8 Iteration and Prototyping

First designs are never perfect.

Process : Design → Evaluate → Improve → Repeat

Formative vs. summative evaluation

Formative evaluation — Evaluation **intended to improve designs** (done **during development**)

Summative evaluation — Evaluation performed **at the end to verify whether the product is good enough.** (done **after development**)

Evaluation Approaches

1. **Expert Evaluation:** **Experts check against guidelines.**
2. **User Evaluation:** **Real users test the system.**

The Hill-Climbing Pitfall

- **Repeatedly making improvements** to reach a better design.
- **start somewhere, evaluate, change to improve, repeat until it cannot get better.**

Problem: Local Maximum

- **Local Maximum:** **Best nearby solution.**
- **Global Maximum:** **Best overall solution.**

Risk: A bad starting design may only become a better version of a bad design.

Two Requirements for Prototyping to Work

1. Know What Is Wrong: Identify problems accurately & how to improve.

2. Good Starting Point: needed to avoid local maxima